

Department of Computer Science & Engineering
Global Campus, Jakkasandra Post, Kanakapura Taluk, Ramanagara District, Pin Code: 562 112

PROJECT CENTRIC LEARNING (PCL) PHASE 2

			PROJECT TEAM NO.	39
PROJECT TITLE		Malicious Insider Detection System in an Organization		
TEAM MEMBERS				
S. No.	USN No.	NAME OF THE STUDENT	CLASS	MOBILE NO.
1	23BTRCL167	A JAVEED BASHA	AIML-A	9059424642
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DETAILS OF THE PROJECT GUIDE	
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DOMAIN OF THE PROJECT	
Cybersecurity Behavioural Analytics	
Scope of the Project	
1. The system monitors high-resolution system telemetry, such as login sequences and file-system access density, to establish what "normal" looks like for every user.	
2. It employs a dual-model approach, using Random Forest for static feature classification and LSTM networks to analyze chronological event sequences.	
3. The framework provides an automated monitoring interface that generates context-rich alerts, allowing security teams to act immediately on suspicious activities	
4. The scope includes a "privacy by design" layer that masks personal identities during analysis to ensure compliance with global data protection standards.	

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PROJECT COORDINATOR

Existing System:-

1. **Perimeter-Centric Defences:** Traditional security is primarily focused on a "fortress" model, using tools like firewalls to stop external adversaries from gaining entry.
2. **Signature-Based Detection:** Existing systems typically look for known "unauthorised entry" signatures rather than monitoring how authorised access is being utilised or potentially abused.
3. **Rule-Based Logic:** Early detection strategies use hard-coded, rigid parameters to flag suspicious activities, such as irregular login times or high-volume data transfers.
4. **Manual Thresholds:** Security protocols often rely on static alarm thresholds that can be easily bypassed by insiders who are familiar with the internal security logic.
5. **High False Positive Rates:** Because these systems lack flexibility and cannot adapt to variable user activities, they frequently generate excessive false alarms, leading to alert fatigue.
6. **Blindness to "Low-and-Slow" Attacks:** Legacy tools are often incapable of identifying malicious patterns that develop gradually over time, as they focus on immediate, overt policy violations.

Problem Identification

- **Perimeter Blindness:** Traditional security tools like firewalls focus on stopping external intruders but are completely blind to threats coming from people who already have valid credentials.
- **Rule Inflexibility:** Legacy systems rely on hard-coded rules that cannot adapt to changing user behaviour, leading to a massive volume of false alarms or missed sophisticated attacks.
- **Exploitation of Authority:** Authorised staff can easily bypass standard security thresholds because they understand the internal logic of the system and know how to avoid triggering simple alerts.

Proposed System:-

1.	Data Acquisition and Pre-processing: The system fuses logs from Active Directory and file system metadata, cleaning the noise via KNN imputation and normalising data using Z-score standardisation: $z = \frac{x - \mu}{\sigma}$
2.	Hybrid Predictive Engine: It combines the classification strength of Random Forests (to catch overt violations) with the temporal sensitivity of LSTM networks (to catch subtle, long-term shifts in behaviour).
3.	Context-Aware Filtering: Unlike generic tools, MIDS adjusts its sensitivity based on organisational roles—recognising that a system administrator's "normal" looks very different from a standard employee's "normal".
4.	Operational Performance: The system achieved a 92% precision rate and a 94% AUC-ROC score in testing, effectively reducing false alerts by 30% while maintaining a low processing latency of 1.8 seconds per sequence.

No. of Modules

Sl. No.	Name of the Module	Remarks
1	Research & Strategic Planning	Involves a comprehensive literature review of insider threat paradigms and defines the detection scope for high-resolution telemetry.
2	Multi-Source Data Acquisition	Focused on the extraction and fusion of authentication logs from Active Directory and sensitive asset file-system metadata
3	Telemetry Preprocessing & Enrichment	Employs KNN imputation for missing values and Z-score standardization to normalize raw data streams for model integrity
4	Hybrid ML Engine Construction	Selection and integration of Random Forest for static feature classification and LSTM networks for temporal behavioral modelling..
5	System Validation & Deployment	Testing the framework against the CERT dataset to assess precision, recall, and real-time monitoring scalability.

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